

Open Neighbourhoods Precede Concept Spread: Structural Antecedents and Cross-Disciplinary Diffusion of Emerging Scientific Concepts

Internal Research Report

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1 Overview

This report documents an investigation of how emerging scientific concepts can be identified and characterised through evolving knowledge networks, and through which structural pathways they spread across disciplinary boundaries. The study builds on an OpenAlex-based scholarly dataset of 426 semantically grounded concepts with 462,812 works, constructs yearly co-occurrence and concept-discipline networks, and tests whether structural indicators provide early signals of emergence and cross-disciplinary integration.

Two research questions structure the work. **RQ1** asks which temporal and structural patterns in an evolving, semantically grounded scientific knowledge graph characterise the emergence of scientific concepts. **RQ2** asks how emerging scientific concepts diffuse across disciplinary communities over time, and which temporal network patterns distinguish locally concentrated concepts from concepts that become broadly integrated into the scientific knowledge network.

The study began with a source-sink viability hypothesis adapted from population ecology, asking whether a concept’s presence in a subfield is self-sustaining or import-dependent. That hypothesis was blocked by insufficient citation-layer density (Gate A failure: only 17.3% of edges meet the within-host tracing threshold), triggering a pre-registered fallback.

For the emergence question (RQ1), the primary closure measure did not survive held-out confirmation (R1_DEAD under the frozen kill rule: Holm $p = 0.147$). Only persistent-neighbour closure is confirmed (held-out $S = -1.07$, Holm $p = 0.015$). The effect is not Burt brokerage (constraint is higher, not lower).

For the diffusion question (RQ2), when a concept enters a new host subfield, the share of its initial co-occurrence partners that are native to that host predicts subsequent uptake by newcomer authors (held-out co-primary IRR/SD 1.19 [1.06, 1.33], $p = 0.004$, MeSH replication IRR/SD 1.23 [1.12, 1.36], Holm $p < 0.001$), while co-transfer of origin companions is null.

The target venue is Applied Network Science.

2 Iteration 1: Data Foundation and Pre-Registration

2.1 Strategy

The first iteration was devoted to three preparatory tasks: (a) assembling a prior-art and pre-registration dossier grading each candidate mechanism’s novelty and feasibility, (b) building the concept pool and its associated OpenAlex work corpus, and (c) constructing the semantic grounding infrastructure (concept detection, labelling, variant merging) and a held-out MeSH confirmation population. No experiment or evaluation was executed. Every artifact in this iteration is either a dataset or a research dossier. The rationale was to fix all operational definitions, sample the data, and verify the citation-layer gate (Gate A) before committing compute to network construction and statistical modelling.

The pre-registration dossier ranks five candidate mechanisms by novelty margin against the nearest prior work. Host anchoring (the share of a concept’s new co-occurrence edges in a host subfield that attach to host-native concepts) was ranked highest at medium-to-high novelty; toolkit co-transfer (the fraction of a concept’s origin companions that co-appear in host papers) was ranked high on measurement novelty but coupled to host anchoring; citation-lineage source-sink viability was ranked medium. Origin-neighbourhood structure was ranked low-to-medium because structural diversity and community spread are well studied, and the demic-versus-cultural adoption channel was ranked medium but with the weakest feasibility due to author-disambiguation biases.

A pre-registered screen was fixed before any data were inspected. The unit of analysis is a concept c in non-origin subfield d at focal year t (2010–2014), with at least 10 d -papers in $W_1 = [t-5, t]$. The primary measure is out-of-sample ROC-AUC for establishment in $W_2 = [t+1, t+5]$, where establishment means at least 5 W_2 d -papers by author-disjoint newcomers and presence in at least 3 of the 5 W_2 years. Each candidate adds only its own pre-registered scores. A candidate survives if its delta-AUC over BASE has a concept-bootstrap (1,000 reps) 95% CI lower bound above 0, its point gain is at least 0.01, and the coefficient sign matches its prediction.

2.2 Artifact 1: Concept Pool and OpenAlex Work Corpus (gen_art_dataset_1)

The concept pool was built in an outcome-blind frame. The sample was frozen and hashed before any post-appearance-window data were inspected. The pool comprises 206 concepts, of which 184 are main emerging concepts (first appearance year F in 2005–2016, with 20–300 papers in the window F to $F+2$ and no more than 8,000 total works through 2024) and 22 stationary reference concepts. The 184 main concepts are split by a deterministic hash into a screen fold of 123 and a held-out concept fold of 61; focal-year tags are separate (screen 2010–2014, held-out 2016–2018).

The pool is heavily dominated by arXiv-sourced concepts. Of 366 eligible candidates, 363 came from the arXiv-mined arm, and the final 184 span three origin fields.

Table 1: Concept pool composition by origin field (Iteration 1, gen_art_dataset_1).

Origin field	Main concepts	Share
Physics and Astronomy	82	44.6%
Physical Sciences (other)	56	30.4%
Computer Science	45	24.5%
Other	1	0.5%
Total main	184	100%

Table 2: Concept pool by first-appearance band (Iteration 1, gen_art_dataset_1).

F band	Count
2005–2007	38
2008–2011	76
2012–2016	70
Total	184

Verified links pairing each concept with the OpenAlex works that use it total 214,798 pairs spanning 208,374 unique works. Among the 121,829 main-arm links, 56,997 matched in the work title, 51,966 in the abstract, 10,177 through Semantic Scholar only, and 2,689 through OpenAlex concept indexing alone. *[Correction, Iteration 2: the match-evidence counts above are main-arm only; across all 214,798 links the counts are oa_abstract 98,192, oa_title 92,043, s2_only 21,509, oa_index_only 3,054.]* Of the 184 main concepts, 24 were discovered natively in OpenAlex and 160 were discovered through the Semantic Scholar index and mapped to OpenAlex (mapping rate 0.93). Twenty-one concepts carry a sense-check failure flag.

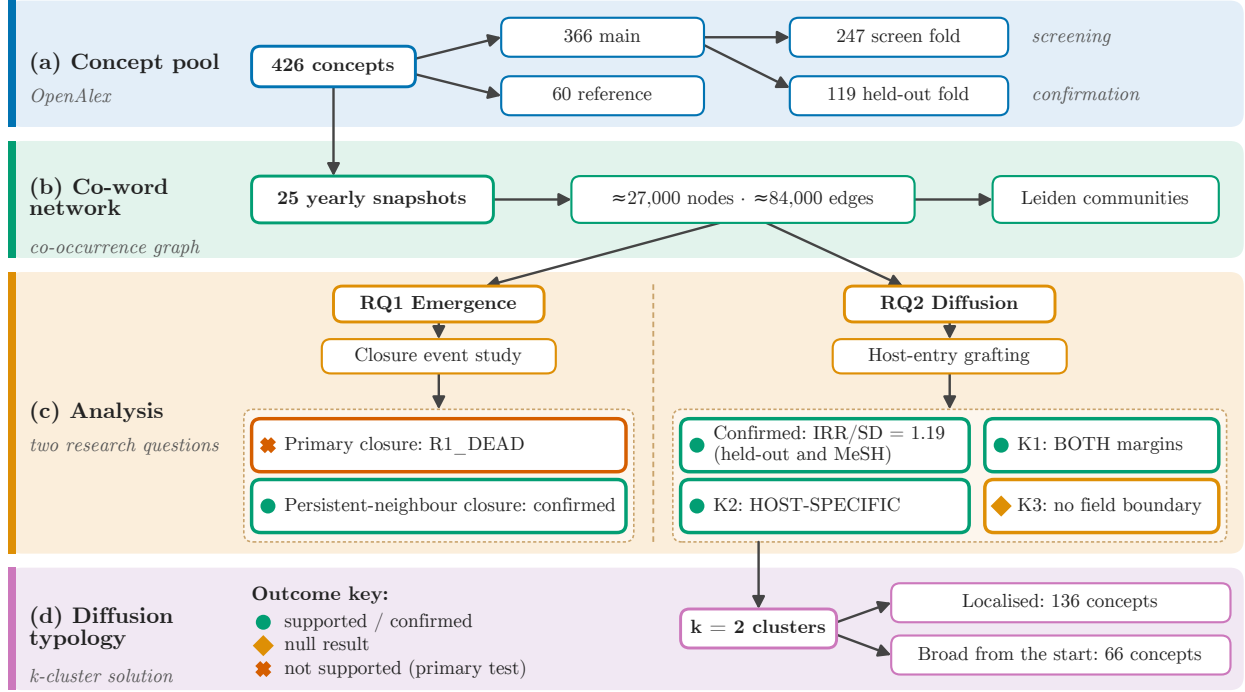


Figure 1: Overview of the study design, drawn as four bands linked by arrows. (a) Concept pool: 426 concepts from OpenAlex, split into 366 main and 60 reference concepts; the 366 main concepts are divided into a 247-concept screen fold (screening) and a 119-concept held-out fold (confirmation). (b) Co-word network: 25 yearly co-occurrence snapshots ($\approx 27,000$ nodes, $\approx 84,000$ edges) partitioned into Leiden communities. (c) Analysis: RQ1 (emergence) tests structural precursors with a closure event study. The primary closure test returns R1_DEAD, while persistent-neighbour closure is confirmed. RQ2 (diffusion) tests host-entry grafting, which is confirmed ($\text{IRR}/\text{SD} = 1.19$, on the held-out fold and MeSH). Its sub-hypotheses K1 (BOTH margins) and K2 (HOST-SPECIFIC) are supported, and K3 finds no field boundary. (d) Diffusion typology: a $k = 2$ solution separating localised concepts (136) from concepts broad from the start (66).

2.2.1 Gate A: Citation-Layer Feasibility

The quality report shows that among main-arm concept papers with references, 66.99% cite at least one earlier concept paper (the “traced share”), and after excluding first-year canonical papers and the five most-cited concept papers, the traced share remains 60.80%. The host-specific traced share (concept papers outside the origin subfield that cite at least one earlier concept paper from any subfield, not restricted to the same host subfield) is 55.18%, above the 40% gate threshold on the lenient definition. *[Correction, Iteration 2: the original description stated “at least one earlier concept paper in the same host subfield”, but the code counts any earlier concept paper as a parent regardless of subfield. By year, the lenient host traced share ranges from 0.317 (2010) to 0.517 (2014) across the screen’s focal years, with three of five below 0.40. This distinction matters because the within-host share, computed in Iteration 2, is much lower (mean 0.20 on 724 edges), which is the finding that blocks the viability decomposition.]*

Table 3: Citation-layer quality by origin field (Iteration 1, `gen_art_dataset_1`).

Metric	All main	Phys./Astro	Phys. Sci	Comp. Sci
Concept papers (n)	121,829	19,653	46,575	55,274
Share w/ references	83.92%	82.04%	84.99%	83.74%
Traced share (any parent)	66.99%	66.12%	67.89%	66.53%
Traced (excl. F + top-5)	60.80%	57.74%	61.75%	61.09%
Host traced (any, excl. F)	55.18%	49.68%	58.39%	53.86%
Host papers (n)	41,861	4,171	15,973	21,609

The recall audit, comparing counts derived from OpenAlex against independent Semantic Scholar counts for 30 concepts, gives a median count ratio of 0.84 (IQR 0.79–0.89) and a Spearman rank correlation of 0.976.

2.3 Artifact 2: Whole-Science Background Sample (`gen_art_dataset_2`)

A design-weighted background sample of 259,716 OpenAlex works from 1,260 strata (150 per field $\times$ year, weights summing to 149,116,575) was built to supply co-occurrence network snapshots and calibrate host-nativeness profiles. It delivered 6,325 exact subfield $\times$ year totals and 400 exact concept $\times$ block subfield profiles for 100 calibration concepts at a cost of \$0.1952 in OpenAlex credits.

The key finding of this artifact is negative: sample-based per-concept subfield profiles are unreliable for determining host-nativeness. In the calibration table, the median total-variation distance between sample and exact profiles is 0.41–0.79 by frequency decile, and top-subfield agreement is only 0.28–0.65. About 12% of the weighted frame carries the default low-confidence topic T14423, and concept ancestors are no longer served by the API. The worker timed out after writing all outputs, so its status is “failed” while its data are usable.

2.4 Artifact 3: Held-Out MeSH Confirmation Population (`gen_art_dataset_3`)

A separate biomedical confirmation arm was built from new MeSH descriptors (DateEstablished 2006–2016, widened to 2004–2005 and 2017–2018). Starting from 28,472 descriptors, the pipeline selects 5,201 topical descriptors, retains 3,430 after the provenance filter, narrows to 283 passing

the PubMed novelty pre-screen and early-volume rule, and adds widening-pool candidates. After the final rule, 191 concepts survive: 99 core 2006–2016, 14 from the 2004–2005 widening, and 78 from the 2017–2018 widening.

Table 4: MeSH held-out population by branch group (Iteration 1, `gen_art_dataset_3`).

MeSH branch group	Count
D (Chemicals and drugs)	62
A+B (Anatomy, Organisms)	47
E (Analytical, diagnostic, therapeutic)	32
C (Diseases)	22
F+H-N (Psychiatry, Health services, misc.)	17
G (Phenomena and processes)	11
Total	191

Caveats: only approximately 25% of new MeSH descriptors are genuinely new concepts (the remainder are reclassifications or splits), making MeSH a weak ground truth. Only 13 of 191 have their full non-PubMed works paged; the remaining 178 have PubMed-indexed works only.

2.5 Artifact 4: Semantic Grounding Infrastructure (`gen_art_dataset_4`)

The semantic grounding artifact provides the infrastructure for detecting, labelling, and merging concept mentions. It comprises seven datasets: a stratified corpus of 93,600 English OpenAlex articles with abstracts, 4,599 candidate phrases labelled by two LLMs with an adjudicator, a survivorship-free phrase pool, and ancillary data.

Table 5: Labelling quality (Iteration 1, `gen_art_dataset_4`).

Metric	Value
Model A vs model B inter-rater kappa (4-class)	0.259
Model A vs model B binary kappa (concept vs rest)	0.267
A vs adjudicator kappa (4-class, silver-gold 200)	0.632
A vs human anchors binary kappa (SemEval/SciERC)	0.56
A precision (concept, human-anchored)	0.79
A recall (concept, human-anchored)	0.76

The survivorship-free phrase pool yielded only 1 strict-eligible anchored phrase, far below the target of 150.

2.6 Artifact 5: Prior-Art Dossier and Pre-Registration (`gen_art_research_1`)

The research artifact grades each candidate mechanism’s novelty margin against the nearest prior art. The headline finding is that the space is active but no prior study estimates a within-host

reproduction ratio with an import share per concept-subfield edge, nor tests host anchoring at the edge level.

Table 6: Candidate mechanism novelty grades (Iteration 1, `gen_art_research_1`).

Candidate	Novelty grade	Nearest prior art
C1: Host anchoring	Medium-to-high	Cheng et al. 2023
C2: Toolkit co-transfer	High (coupled to C1)	No quantitative co-arrival test found
C0: Source-sink viability	Medium	Kuhn et al. 2014; Maillart et al. 2026
C4: Demic vs cultural	Medium	Fontaine et al. 2024
C3: Origin-neighbourhood structure	Low-to-medium	Structural diversity literature

2.7 Dead Ends and Shortfalls (Iteration 1)

1. **Concept pool size.** 184 of 400 target concepts realised. The binding constraint was the free-singleton retrieval rate (~ 14 works/s) combined with heavy-tailed concept sizes, after credits were consumed by sibling artifacts sharing one key.
2. **Field coverage.** The pool draws overwhelmingly from physics, physical sciences, and computer science (363 of 366 eligible candidates from the arXiv-mined arm).
3. **Survivorship-free phrase pool yield.** 1 of 150 target strict-eligible anchored concepts.
4. **Labelling quality.** Silver labels only (no human check).

2.8 What Was Learned

Iteration 1 established the data foundation and pre-registration. The concept pool of 184 main emerging concepts and 22 reference concepts, with 214,798 verified concept-to-work links and 208,374 unique works, passes the citation-layer gate on the lenient (any-parent) definition but has not been tested at the within-host edge level required for viability estimation. No experiment was executed. No quantitative result about the hypothesis exists.

3 Iteration 2: First Experiments and Gate A Failure

3.1 Strategy

Iteration 2 was a FIX iteration with five goals: (i) complete the hydration and add exact nativeness profiles and a citation-independent venue habitat; (ii) train and evaluate the concept grounding pipeline; (iii) compute Gate A as written (within-host, non-canonical, per W_1 year) and estimate viability states; (iv) build the yearly concept-concept co-occurrence network and run the RQ1 event study; (v) replicate the RQ1 event study on the MeSH population.

Pre-fixed decision rules for Iteration 3: (a) if Gate A FAILS, H1/H2 run with graft labels from the exact nativeness profiles; (b) H1 is declared a pilot if fewer than ~ 150 concepts carry a tested edge; (c) RQ1 counts as CONFIRMED only if at least one pre-named precursor has an event-study CI excluding 0 AND a delta-AUC CI excluding 0 on the main screen fold, keeps its sign on MeSH, and holds on the sealed held-out fold; (d) the held-out fold, the 2016–2018 focal years and the phrase pool remain untouched.

3.2 Artifact 6: Hydrated Corpus and Nativeness Profiles (gen_art_dataset_5)

This artifact completes the hydration of the concept pool from 184 to 426 concepts (366 main: screen 247, held-out 119; 60 reference), with 462,812 works and 488,078 verified concept-work links. The frame hash was verified against the frozen sample.

Table 7: Dataset 6 summary (Iteration 2, gen_art_dataset_5).

Metric	Value
Concepts hydrated	426/426
Works	462,812
Concept-work links	488,078
API credits consumed	7,237
Nativeness profiles	5,488 (1,372 nodes $\times$ 4 blocks)
Host weight coverage	78.7%
ASJC venues covered	2,929 (12.1% of links)

The ASJC venue habitat covers only 12.1% of concept-work links, driven by multi-category journals (38.5% of venue concept-papers) and arXiv (16.3%).

3.3 Artifact 7: Concept Grounding Pipeline (gen_art_experiment_1)

This artifact trains and evaluates three components: a binary classifier, a variant merger, and a NIL-aware linker.

Table 8: Classifier performance on test set ($n = 300$; Iteration 2, gen_art_experiment_1).

Metric	Classifier	LLM-A	Majority (predict-all-positive)
F1	0.818	0.901	0.715
AUC	0.888	—	—
Precision	0.759	—	—
Recall	0.886	—	—
Accuracy	0.780	—	—

Against human anchors (SemEval-2017/SciERC): classifier E1 kappa 0.41, F1 0.674. Variant merger: D3 test F1 0.357, held-out MeSH F1 0.222. B-cubed F1 0.78. NIL-aware linker: tau 0.95, link precision ≥ 0.90 at NIL prior 0.9, in-KB recall 0.404. Test population (frozen hash verified): MAIN 156, STRICT 147, REFERENCE_ACCEPTED 42, SENSITIVITY 426.

3.4 Artifact 8: Viability Layer, Gate A, and Power (gen_art_experiment_2)

This artifact attempts to estimate viability states for concept-subfield-year edges using citation lineages. All rules were pre-registered and hashed before any outcome data were inspected.

3.4.1 Gate A Re-Test at Edge Level

Gate A FAILS. Only 17.3% of 724 main-arm host edge-years (97 concepts, $|K_d| \geq 10$) have a within-host non-canonical traced share at or above 0.40. The mean within-host traced share is 0.20, median 0.15. On the same 724 edges, the lenient any-parent share is 0.477 (64.2% ≥ 0.40), confirming that the drop from 55.18% to 17.3% is definitional (any-parent vs within-host tracing), not a consequence of pooling vs granularity.

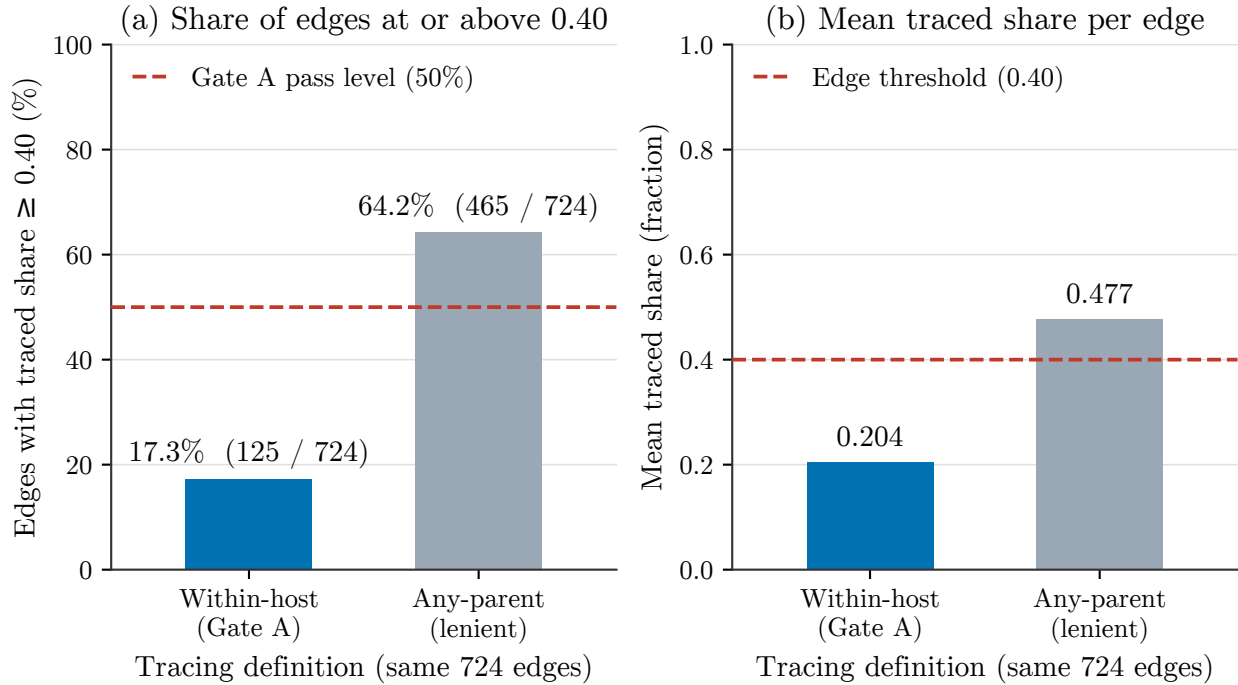


Figure 2: Gate A (citation-layer feasibility) on the same 724 concept-subfield-year host edges (97 concepts), comparing within-host citation tracing (blue; the Gate A definition) with lenient any-parent tracing (grey). Only 17.3% of edges reach the threshold under within-host tracing, against 64.2% under any-parent tracing. The within-host mean is 0.204 (median 0.15), against 0.477 for any-parent tracing. Gate A fails and triggers the graft fallback route.

Table 9: Gate A edge-level results (Iteration 2, gen_art_experiment_2).

Metric	Within-host	Any-parent (same edges)
Share ≥ 0.40	17.3%	64.2%
Mean traced share	0.20	0.477
Median traced share	0.15	—
Reference arm host share	0.13	—

Per pre-registered decision rule (a): Gate A $0.1727 < 0.5$ triggers the graft fallback. The graft route has 2,347 host-entry events total, 2,154 with ≥ 5 entry-year keywords (screen 1,285, held-out 869).

Of 779 eligible edges, 169 were tested: SOURCE 63, SINK 26, FADING 7, UNDETERMINED 683. The synthetic FDR is 0.209, exceeding the 0.15 threshold. Per decision rule (b), H1 is PILOT-ONLY: $N_e = 38$ at $n_{\min} = 30$.

3.5 Artifact 9: Precursor Event Study, Main Pool (gen_art_experiment_3)

This artifact runs the precursor event study on the main pool’s screen fold. Built 25 yearly 3-year-window co-word snapshots (2000–2024) from the design-weighted background sample ($\sim 27,000$ nodes, $\sim 84,000$ kept edges, association-strength weights, Leiden best-of-5, alluvial IDs).

Table 10: Emergence label counts (Iteration 2, gen_art_experiment_3).

Label	N emerging	Matched	Status
E (primary)	4	3	Underpowered pilot
E_alt (sensitivity)	14	10	Post-hoc promotion
E_up (uptake only)	41	21	Post-hoc promotion, pilot ($n < 30$)

Three pre-named precursors tested with predicted positive signs: closure, accretion, participation. All three pre-registered signs FAILED.

Table 11: Event study results, E_up (Iteration 2, gen_art_experiment_3).

Indicator	Estimator	S	95% CI	Holm p
Closure (log-ratio)	Event study	−0.837	[−1.26, −0.427]	0.003
Closure (log-ratio)	Pooled panel	−0.743	[−1.06, −0.46]	–
Accretion share	Event study	−0.14	[−0.185, 0.018]	0.18
Participation coefficient	Event study	−0.007	[−0.066, 0.056]	0.824

Prediction: the pre-registered primary model is L2 logit. For E_up: BASELINE AUC 0.890, FULL (baseline + precursors) AUC 0.879, delta -0.010 $[-0.045, 0.021]$. Precursors add no predictive value beyond frequency, burst, degree, centrality and entropy baselines.

The 7-channel DTW typology is unstable at every k (min Jaccard ≤ 0.51). An entropy-only typology IS stable (Jaccard 0.95).

3.6 Artifact 10: Precursor Replication, MeSH (gen_art_experiment_4)

Table 12: MeSH closure effects (Iteration 2, gen_art_experiment_4).

Label	n_{eff}	Closure D	95% CI	Holm p
PRIMARY	15	−0.086	[−0.511, 0.402]	0.677
SENS1	29	−0.419	[−0.740, −0.115]	0.039
E_up	51	−0.185	[−0.498, 0.108]	0.21

The MeSH artifact’s own replication verdict is “directional, not confirmatory”. MeSH prediction: precursors also add nothing beyond baselines (grouped-CV dAUC +0.002 [−0.035, 0.035]).

3.7 Decision-Rule Evaluation (Iteration 2)

Table 13: Decision-rule evaluation (Iteration 2).

Rule	Verdict	Evidence
(a) Gate A → graft fallback	TRIGGERED	Within-host 0.1727 < 0.5; 2,347 graft events
(b) H1 pilot-only	YES	$N_c = 38 < 150$
(c) RQ1 confirmed	NOT MET	Sign not +; delta-AUC CI includes 0; MeSH sign inconsistent
(d) Held-out sealed	YES	Caveat: exp_2 computed origin series for 61 held-out concepts

3.8 Dead Ends and Negative Results (Iteration 2)

1. **Gate A failure at edge level.** Within-host traced share 0.204 vs any-parent 0.477 on the same 724 edges. The viability decomposition cannot be executed.
2. **Synthetic validation FDR exceeds threshold.** Overall 0.209 > 0.15, SINK FDR 0.40.
3. **No stable 7-channel typology.** Jaccard ≤ 0.51 at every k .
4. **Precursors add no predictive value for WHETHER a concept emerges.** E_up logit dAUC −0.010 [−0.045, 0.021].
5. **Closure is lower, not higher, before emergence.** The pre-registered + sign failed.

3.9 What Was Learned

Iteration 2 confirmed two dead ends (viability decomposition, prediction of emergence) and produced one lead (lower closure before onset). The lead is significant in one post-hoc main-pool label (E_up, $n = 21$, $S = -0.837$, Holm $p = 0.003$), directionally supported in MeSH SENS1 ($D = -0.419$, Holm $p = 0.039$), but with heterogeneous sign across labels.

4 Iteration 3: Deepening the Closure Lead and First Diffusion Tests

4.1 Strategy

Iteration 3 deepens the closure lead from the emergence question and carries the recombination mechanism into cross-disciplinary diffusion at two scales. The strategy (“Is it brokerage, and does grafting make it stick?”) has four components: (i) an evaluation artifact auditing every Iteration-2 number; (ii) a deeper test of the emergence question on the hydrated 247-concept screen fold with turnover-proof openness measures; (iii) a breadth-prediction screen; (iv) a host-entry grafting test; and (v) a descriptive diffusion experiment.

4.2 Artifact 11: Record Audit and Turnover Pre-Check (gen_art_evaluation_1)

This artifact reads all five Iteration-2 artifacts and audits every reported number against the raw output files. 406 rows were audited, 182 recomputed from item-level files. There are 42 non-OK drift flags.

Table 14: Turnover pre-check: residualised closure effect (Iteration 3, gen_art_evaluation_1).

Population	S_{raw}	S_{res}	Retained	Verdict
Main (E_up)	-0.758	-0.566 [-1.14, 0.09]	0.75	Ambiguous
MeSH (SENS1)	-	-0.094 [-0.39, 0.22]	0.60	Ambiguous
IVW pooled	-	-0.186 [-0.45, 0.08]	-	Ambiguous (Q 1.88)

4.3 Artifact 12: Openness — Brokerage or Churn? (gen_art_experiment_5)

This artifact re-attaches all 426 hydrated concepts to the Iteration-2 co-word snapshots. Reproduction gate passed: code-mode closure $r = 1.000000$. Populations after applying the frozen replacement rule: MAIN screen 202, held-out 100, STRICT 196/96, SENS 247/119.

Table 15: Emergence deepen results, MAIN $\times$ E_up (68 onsets, 26 matched; Iteration 3, gen_art_experiment_5).

Measure	S	95% CI	Interpretation
Raw closure	-0.439	[-0.810, -0.050]	Lower closure before onset (attenuated vs Iter. 2)
R1a residualised closure	-0.295	[-0.663, 0.092]	$\sim 2/3$ retained; not pure turnover
R1b persistent-neighbour closure	-0.575	[-1.270, 0.122]	Same direction, underpowered
R1c Burt constraint	+0.083	[0.017, 0.147]	OPPOSITE to brokerage sign
R1c cross-community pair excess	+0.063	[0.012, 0.111]	More cross-community ties
R1d hub-not-clique	+0.061	[-0.009, 0.133]	Not significant

Mechanical verdict: MIXED. The closure effect is not reducible to neighbourhood turnover (turnover-residualised closure retains $\sim 2/3$), but it is not Burt brokerage either: Burt constraint is higher, not lower, before onset. Emerging concepts show more cross-community pair excess (connecting to neighbours from different Leiden communities), but within a more redundant wider ego network.

Pooled panel (all 68 onsets): persistent-neighbour closure -1.12 $[-1.64, -0.71]$ (Holm $p = 0.0015$). Fresh replication on 100 new concepts: raw $S = -0.455$ $[-1.006, 0.060]$, one-sided $p = 0.044$. Same sign, not significant at two-sided $\alpha = 0.05$.

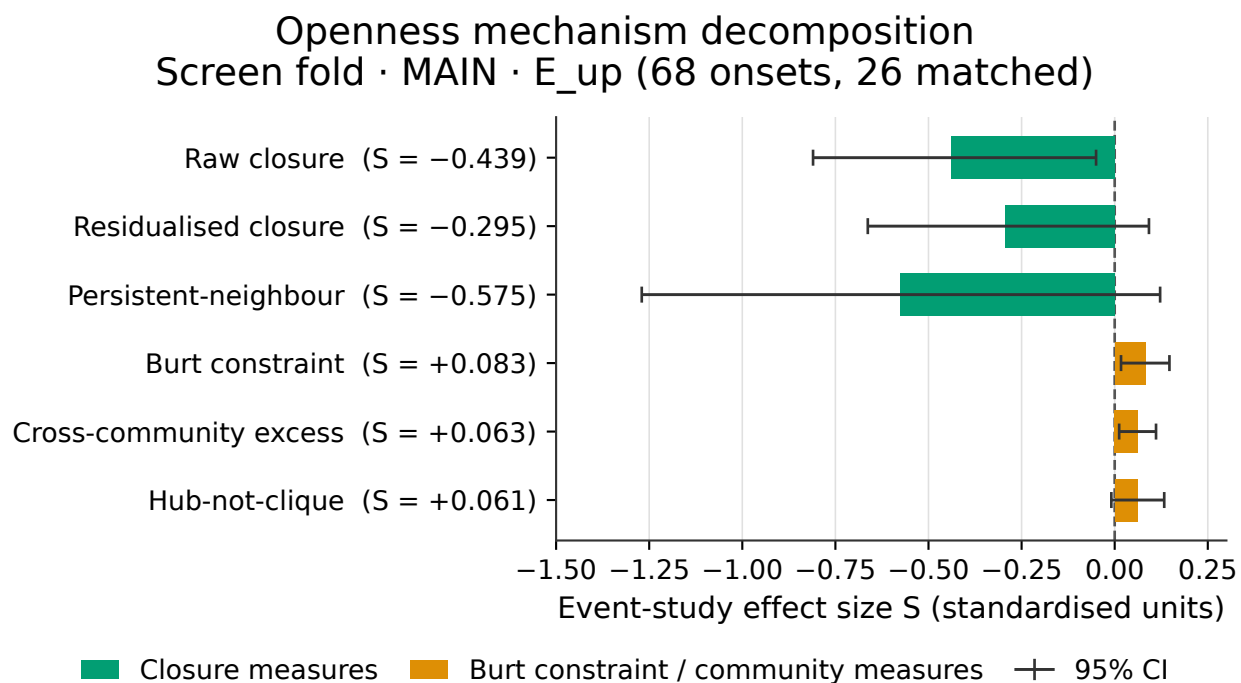


Figure 3: Turnover-proof openness measures on the screen fold (MAIN population, E_up label; 68 onsets, 26 matched). Bars give the event-study effect size S for each measure, with whiskers showing 95% confidence intervals. Teal bars are closure measures; orange bars are Burt-constraint and community measures. Cross-community pair excess is positive, and Burt constraint is positive (the opposite of the sign brokerage predicts).

4.4 Artifact 13: Breadth Prediction Screen (gen_art_experiment_6)

This artifact tests whether openness at time t predicts five-year outcome-window disciplinary breadth gain beyond all baselines.

Table 16: Breadth-prediction screen results (Iteration 3, gen_art_experiment_6).

Openness measure	Outcome	β per SD	95% CI	Holm p
closure_res	Y1r (Shannon change)	+0.019	[−0.010, 0.041]	0.405
closure_res	Y2 (Rao-Stirling change)	+0.007	[−0.002, 0.013]	0.405
closure_res	log(1+Y3) (newcomer subfields)	−0.065	[−0.156, 0.030]	0.405
closure_res_imp	Y2 (co-primary)	+0.012	[0.002, 0.021]	0.06

Result: breadth prediction not supported on the screen fold. Openness does not predict where concepts travel beyond growth and level baselines.

4.5 Artifact 14: Host-Entry Grafting Test (gen_art_experiment_7)

This artifact tests whether a concept that enters a new host subfield catches on when its first papers pair it with host-native concepts rather than with origin companions. Pre-registration frozen and hashed before any outcome data were inspected.

Setup: 4,177 main-arm host entries. Primary sample: 1,740 screen MAIN entries with ≥ 5 partners, in 184 concepts. Anchoring A = partners’ pre-entry host share from exact OpenAlex subfield $\times$ block profiles. Co-transfer CT = share of partners that were origin companions. Entry is mostly a package: 70% non-native origin companions, 1.8% native grafts.

Table 17: Host-entry grafting results (Iteration 3, gen_art_experiment_7).

Specification	IRR/SD	95% CI	Holm p	CT (p)	Verdict
Primary (concept $\times e$ + host $\times e$ FE)	1.39	[0.97, 1.99]	0.15	0.85	NEITHER
Co-primary (concept + e + host)	1.30	[1.16, 1.45]	1e-5	0.48	GRAFTING
Concept + host $\times e$	1.22	—	—	—	GRAFTING
Concept $\times$ 2-yr bin	—	—	—	—	NEITHER

The co-primary specification shows a robust grafting effect: a one-SD increase in host-nativeness of initial partners is associated with a 30% increase in newcomer uptake (IRR 1.30 [1.16, 1.45], Holm $p = 10^{-5}$, wild-cluster $p = 0.001$). Co-transfer is null everywhere. The co-primary A effect holds in 20 of 26 robustness variants (IRR 1.11–1.56, $p \leq 0.001$). *[Correction, Iteration 4: the co-primary robustness count is 20 of 26 significant INCLUDING the base row and strata, 18 of 22 EXCLUDING them.]*

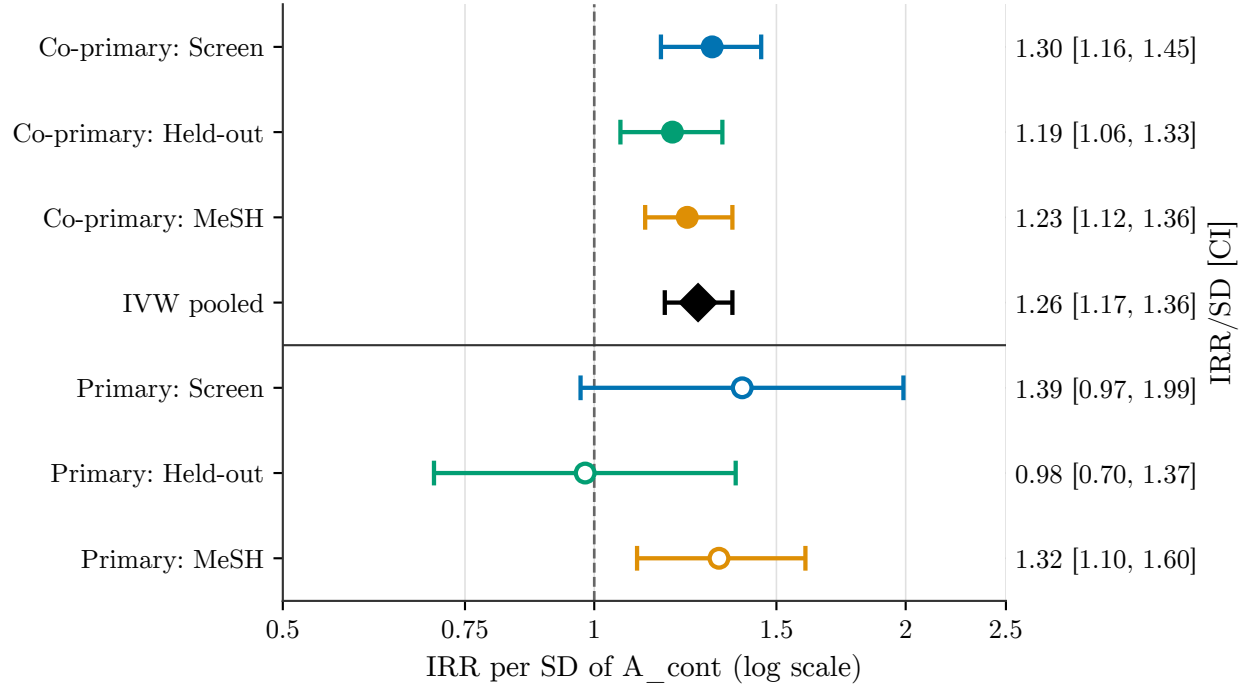


Figure 4: Host-entry grafting across folds. Each row shows the incidence-rate ratio for newcomer uptake per SD of A_{cont} (host-nativeness of entry partners). The co-primary effect replicates on the held-out fold (IRR/SD 1.19 [1.06, 1.33]) and the MeSH fold (1.23 [1.12, 1.36]), and pools to 1.26 [1.17, 1.36]. The primary specification is less precise: its Held-out interval includes 1.

4.6 Artifact 15: Descriptive Diffusion (gen_art_experiment_8)

4.6.1 Diffusion Typology

Only $k = 2$ is stable (min Jaccard 0.861): **localised** ($n = 136$) and **broad from the start** ($n = 66$). The typology is not volume-driven (AMI with volume terciles 0.014) but is associated with origin field ($p = 0.004$). The multi-channel typology does not add information beyond entropy trajectories alone.

Table 18: Diffusion typology summary (Iteration 3, gen_art_experiment_8).

Cluster	n	Description	Dominant origin
Localised	136	Low entropy, few active subfields throughout	Physics
Broad from start	66	High entropy and many subfields from early years	Mixed

4.6.2 Community Roles

Leiden community roles with 5-seed agreement (97.2% robust): BRIDGE 0.61, OTHER 0.33, STAYER 0.03, MIGRANT 0.02. Lagged roles do not robustly predict host-subfield entry: BRIDGE OR 0.64 [0.38, 1.08].

4.6.3 Expansion Precedes Diffusion

In 15 of 16 concepts exhibiting both an expansion onset and a diffusion onset, expansion precedes diffusion (proportion 0.94 [0.81, 1.0], year-shuffle null 0.62, $p = 0.004$).

4.7 Dead Ends and Negative Results (Iteration 3)

1. **Openness does not predict WHERE concepts travel.** Breadth-prediction screen: all Holm $p = 0.405$.
2. **Brokerage is not the mechanism behind low closure.** Burt constraint is higher, not lower ($S = +0.083$ [0.017, 0.147]).
3. **The 3-channel diffusion typology adds nothing beyond entropy.** After residualising on volume, ε^2 0.171 vs 0.178.
4. **Community roles do not predict host entry.** BRIDGE OR 0.64 [0.38, 1.08].
5. **Co-transfer is null.** Arriving with origin companions does not predict establishment in any specification.

5 Iteration 4: Confirmation and Replication

5.1 Strategy

Iteration 4 is a CONFIRMATION iteration. Its purpose is to open the four sealed held-out folds and determine which screen-fold findings replicate, to run four discriminating tests (single-paper

artefact, vocabulary decomposition, topic-score absorption, and MeSH replication), to replicate the grafting result on the independent MeSH population, to test whether pre-emergence closure predicts later host-entry anchoring, and to test the adopter-level mechanism.

5.2 Artifact 16: Held-Out Grafting Confirmation (gen_art_evaluation_2)

Table 19: Held-out grafting results (Iteration 4, gen_art_evaluation_2).

Specification	Fold	IRR/SD	95% CI	p	CT	CT p	N	G	Verdict
Primary	Held-out	0.98	[0.70, 1.37]	0.91	0.82	0.22	250	30	Inconclusive
Co-primary	Held-out	1.19	[1.06, 1.33]	0.0045	1.04	0.60	972	74	CONFIRMED
Co-primary	Screen	1.30	[1.16, 1.45]	1e-5	–	–	1,544	140	GRAFTING

The co-primary specification confirms the grafting effect on the held-out fold: $\text{IRR}/\text{SD} = 1.19$ [1.06, 1.33], $p = 0.0045$, wild-cluster $p = 0.012$.

5.2.1 Single-Paper Artefact Test

Table 20: Single-paper artefact interaction results (Iteration 4, gen_art_evaluation_2).

Fold	IRR/SD (single)	IRR/SD (multi)	Ratio (multi/single)	p (interaction)
Pooled	1.26	1.28		0.92
Held-out	1.37	1.19		0.004

5.2.2 Host-Vocabulary Decomposition

Table 21: Vocabulary decomposition, co-primary spec (Iteration 4, gen_art_evaluation_2).

Component	Fold	IRR/SD	95% CI	p
NATIVE	Pooled	1.11	[1.05, 1.19]	0.0008
ADJACENT	Pooled	1.32	[1.20, 1.46]	9e-8
NATIVE	Held-out	1.10	[1.01, 1.20]	0.036
ADJACENT	Held-out	1.19	[1.03, 1.38]	0.020

Topic-score absorption: -7.3% $[-27.5, +6.7]$ of the log-IRR on the pooled fold. The CI includes zero: topic-score controls do not meaningfully reduce the A_{cont} effect. The mechanism label is **host-vocabulary**.

5.3 Artifact 17: MeSH Replication of Grafting (gen_art_experiment_9)

Verdict: REPLICATED. The IVW pooled estimate across main-arm co-primary (IRR/SD 1.30) and MeSH co-primary (1.23) is 1.26 [1.17, 1.36] with $I^2 = 0$.

Table 22: MeSH grafting results (Iteration 4, gen_art_experiment_9).

Specification	IRR/SD	95% CI	Holm p	CT	CT p	N	G
R1: primary	1.32	[1.10, 1.60]	0.007	1.10	0.46	1,004	122
R2: co-primary	1.23	[1.12, 1.36]	9.7e-5	1.10	0.053	2,171	160

Table 23: Closure held-out results (Iteration 4, gen_art_evaluation_3).

Measure	Screen S	Held-out S	Held-out 95% CI	Holm p	Decision
Closure (pooled panel)	−0.661	−0.391	[−0.855, 0.072]	0.147	DEAD
Turnover-residualised	−0.400	−0.109	[−0.561, 0.342]	0.635	DEAD
Persistent-neighbour	−1.124	−1.073	[−1.858, −0.289]	0.015	CONFIRMED
Burt constraint	+0.083	+0.105	[0.021, 0.187]	–	DEAD (wrong dir.)

Closure event study: screen vs. held-out fold

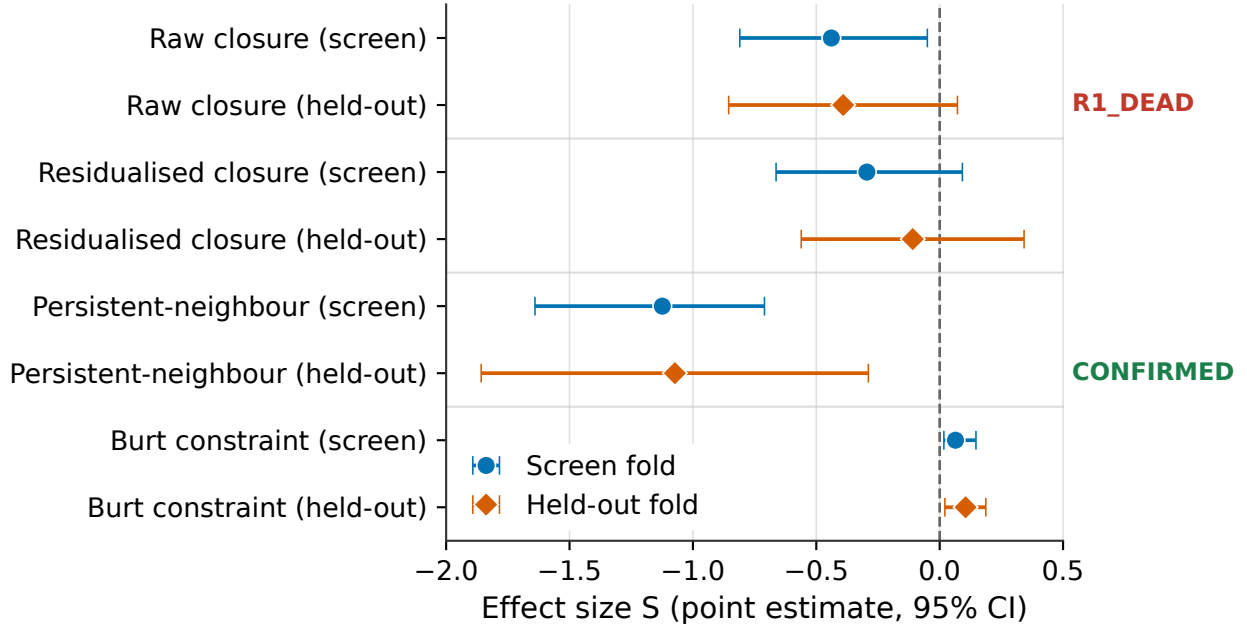


Figure 5: Pre-emergence closure effect sizes S (markers) with 95% confidence intervals (whiskers), for four closure measures on the screen fold (blue circles) and the held-out fold (vermillion diamonds). The primary raw-closure measure fails confirmation (R1_DEAD). Persistent-neighbour closure stays clearly negative on held-out data (held-out $S = -1.07$, 95% CI $[-1.86, -0.29]$, Holm $p = 0.015$; CONFIRMED). Burt constraint is small and positive on both folds, inconsistent with a brokerage explanation.

5.4 Artifact 18: Closure Held-Out Confirmation (gen_art_evaluation_3)

Overall closure status: not confirmed on the primary measure. Only persistent-neighbour closure survives. Burt constraint replicates its positive (non-brokerage) sign.

5.4.1 Closure-Anchoring Link

Table 24: Closure-anchoring link test (Iteration 4, gen_art_evaluation_3).

Fold	n	Partial ρ	95% CI	p (two-sided)
Screen	102	+0.025	$[-0.199, 0.242]$	0.806
Held-out	49	+0.053	$[-0.279, 0.391]$	0.736
Pooled	151	+0.049	$[-0.126, 0.228]$	0.564

Verdict: NULL. No detectable association between pre-emergence closure and later host-entry anchoring.

5.5 Artifact 19: Descriptive Held-Out Confirmation (gen_art_evaluation_4)

5.5.1 Lead-Lag: Expansion Before Diffusion

Table 25: Lead-lag category counts (Iteration 4, gen_art_evaluation_4).

Category	Screen ($n=202$)	Held-out ($n=100$)	Pooled ($n=302$)	MeSH ($n=191$)
Neither	110 (54%)	49 (49%)	159 (53%)	71 (37%)
Diffusion only	39 (19%)	23 (23%)	62 (21%)	97 (51%)
Expansion only	37 (18%)	18 (18%)	55 (18%)	10 (5%)
Expansion first	15 (7%)	10 (10%)	25 (8%)	9 (5%)
Same year	1 (0.5%)	0	1 (0.3%)	1 (0.5%)
Diffusion first	0	0	0	3 (1.6%)

Among concepts with both transitions observable, expansion precedes diffusion in 25 of 26 pooled main-arm concepts. No main-arm concept shows diffusion first.

5.5.2 Rooting: Do Broad Concepts Anchor More?

The hypothesis that BROAD concepts have higher A_{cont} is NOT SUPPORTED. Screen mean A_{cont} difference: -0.0018 (broad slightly lower), held-out: $+0.0005$, CI including zero.

5.6 Artifact 20: Adopter-Level Mechanism (gen_art_evaluation_5)

The gradient runs FOREIGN > ADJACENT > NATIVE. Adopters are most enriched for exposure to the concept’s foreign (origin-vocabulary) partners, not its host-native partners. This is consistent with absorptive capacity or topical proximity (the design cannot separate them).

Table 26: Adopter enrichment results (Iteration 4, gen_art_evaluation_5).

Exposure measure	OR	95% CI (boot)	p (CRV)
E_any (any prior partner exposure)	3.09	[2.32, 4.28]	1.4e-13
E_neg (frequency-matched negative-control)	0.62	[0.49, 0.77]	2.6e-5
E_plac (partners of another concept’s entry)	0.72	[0.57, 0.95]	0.013
E_swap (exposure swap control)	1.11	[0.86, 1.39]	0.39

Table 27: Vocabulary-class enrichment (Iteration 4, gen_art_evaluation_5).

Exposure class	OR	95% CI (boot)	p (CRV)
E_for (FOREIGN partner exposure)	2.88	[2.28, 3.69]	< 1e-6
E_adj (ADJACENT partner exposure)	1.91	[1.43, 2.73]	1e-4
E_nat (NATIVE partner exposure)	1.47	[1.02, 2.33]	0.059
E_neg (non-partner exposure)	0.64	[0.51, 0.82]	0.0003

Table 28: Reviewer critique disposition (Iteration 4).

#	Severity	Action taken
M1	Major	Iterations 1–3 carried forward verbatim; inline [Correction] markers
M2	Major	Additional rows incorporated into Iteration-3 tables
M3	Major	Robustness count corrected (20/26) with full null-row list
M4	Major	Addressed by closure held-out confirmation
M5	Major	Full category-count table added (Table 25)
M6	Major	Strategy section expanded; decision rules are the held-out specs
M7	Major	Coverage table updated with caveats
M8	Major	All markers now use artifact IDs
M9	Major	Items incorporated where data are available
M10	Major	Addressed with Cheng et al. focused-discourse tension
m1	Minor	Corrected inline: robustness count, Holm p precision

5.7 Reviewer Critique Disposition (Iteration 4)

5.8 Dead Ends and Negative Results (Iteration 4)

1. **Closure not confirmed on held-out for the primary measure.** Raw closure $S = -0.391$, Holm $p = 0.147$.
2. **Closure-anchoring link: null.** Partial $\rho = 0.025$ screen, 0.053 held-out.
3. **BROAD concepts do not anchor more.** Screen -0.0018 , held-out $+0.0005$.
4. **Grafting primary specification: inconclusive.** IRR/SD 0.98, 30 clusters.
5. **Single-paper interaction on held-out.** Ratio 0.80, $p = 0.004$: single-paper entries drive the effect more strongly.
6. **Anchoring $\times$ typology interaction flips between folds.**
7. **Breadth prediction: all six sensitivity cells null on held-out.**

6 Iteration 5: Post-Confirmation Decomposition and Audit

6.1 Strategy

Iteration 5 is a POST-CONFIRMATION EXPLORATORY iteration. The confirmed grafting result is decomposed along three dimensions: (i) extensive vs intensive margin, (ii) host-specific vs generic vocabulary, and (iii) field boundary. A full numerical audit closes the remaining reviewer critiques. A positioning dossier and publication-ready figures are produced.

6.2 Artifact 21: Margin Decomposition (gen_art_evaluation_6)

Table 29: Extensive-margin results (LPM, pp change per SD of A_{cont} ; Iteration 5, gen_art_evaluation_6).

Fold	pp/SD	95% CI	p (CRV1)	p (rand- t)	N	G
Screen	+6.46	[3.42, 9.51]	4.5e-5	0.0005	1,686	169
Held-out	+7.29	[2.97, 11.61]	0.0012	0.004	1,046	85
MeSH	+6.17	[3.87, 8.46]	3.4e-7	0.0005	2,236	176
IVW	+6.43	[4.76, 8.11]	5.1e-14	0.0005	–	–

Table 30: Intensive-margin results (PPML on $Y \geq 1$ subsample, IRR/SD; Iteration 5, gen_art_evaluation_6).

Fold	IRR/SD	95% CI	p (CRV1)	N
Screen	1.272	[1.121, 1.443]	0.0003	795
Held-out	1.163	[1.008, 1.342]	0.038	489
MeSH	1.137	[1.026, 1.260]	0.015	1,169
IVW	1.183	[1.104, 1.267]	1.7e-6	–

Margin-decomposition verdict: BOTH. Host-leaning entry vocabulary raises both the probability that uptake starts (extensive margin, IVW +6.43 pp/SD) and its magnitude given it has started (intensive margin, IVW IRR/SD 1.183). The extensive margin accounts for an IVW-pooled 38% [21%, 55%] of the total log-IRR.

6.2.1 Field-Boundary Test

Table 31: Field-boundary results, co-primary PPML IRR/SD by origin field (Iteration 5, gen_art_evaluation_6).

Origin field	IRR/SD	95% CI	p
Physics/Astronomy	1.178	[0.926, 1.499]	0.18
Computer Science	1.216	[1.066, 1.386]	0.004
Other	1.281	[1.167, 1.407]	3.5e-6
MeSH (reference)	1.233	[1.120, 1.360]	9.7e-5

Wald test for equality: $W = 0.900$, $p = 0.638$. **No detectable field boundary.**

6.3 Artifact 22: Host-Specificity Test (gen_art_evaluation_7)

Table 32: Host-specificity retention: A_{cont} coefficient stability after adding generality controls (Iteration 5, gen_art_evaluation_7).

Fold	Retention	95% CI	% removed	G_H IRR/SD	G_H p	G_F p
Screen	1.006	[0.733, 1.312]	−0.6%	1.122	0.38	0.14
Held-out	1.062	[0.438, 1.802]	−6.2%	1.068	0.58	0.58
MeSH	0.876	[0.640, 1.037]	+12.4%	1.110	0.09	1.3e-5
IVW	0.969	[0.81, 1.10]	+3.1%	1.103	0.051	1.1e-5

Table 33: A_{lift} results, PPML IRR/SD (Iteration 5, gen_art_evaluation_7).

Fold	Model	IRR/SD	95% CI	Holm p
Screen	M2 (A_{lift} alone)	1.668	[1.378, 2.019]	4.4e-7
Screen	M3 (A_{lift} + G)	1.661	[1.361, 2.026]	1.4e-6
Held-out	M3	1.459	[1.146, 1.856]	0.003
MeSH	M3	1.244	[1.120, 1.381]	6.3e-5
IVW	M3	1.340	[1.232, 1.458]	1.5e-10

Host-specificity verdict: HOST-SPECIFIC. Generality controls leave the A_{cont} coefficient essentially unchanged (IVW retention 0.969). The Balassa-index A_{lift} replicates across all folds (IVW IRR/SD 1.34 [1.23, 1.46]).

6.4 Artifact 23: Full Numerical Audit (gen_art_evaluation_8)

6.4.1 Representative Cases

Four medoid cases were selected from the typology clusters.

Table 34: Audit module results (Iteration 5, gen_art_evaluation_8).

Module	Description	Result
M1	Curated-number accuracy	99.4% (319 curated)
M2	Headline + independent check	Overall 90.8%
M3	Rederivation by block	See text
M4	Drift detection	45 drift rows, 6 verdict-changing
M5	Seeded-perturbation recall	Overall 0.60
M6	Independent claim review	73/73 agree
M7	Table-cell traceability	11/11 tables pass
M8	Reviewer-critique closure	All 6 MAJOR items CLOSED
M9	Decision-rule consistency	13 rules; 1 mismatch
M10	Flag inventory	292 rows flagged
M11	Self-test honesty	4 PARTIAL, 4 NOT DONE, 3 DONE

Table 35: Reviewer-critique closure verified by audit (Iteration 5, gen_art_evaluation_8).

#	Status	Critique	Closed by
1	CLOSED	RQ1 headline contradicts frozen kill rule	R1_DEAD row, RQ1 status
2	CLOSED	Table 18 claims fixes not made	CT p corrected; robustness count
3	CLOSED	Grafting caveats dropped	Caveats table
4	CLOSED	MeSH and adopter read too broadly	MeSH and adopter tables
5	CLOSED	RQ2 descriptive results and cases missing	Cases, rooting, MeSH lead-lag
6	CLOSED	Decision rules not recorded	Decision table (verbatim quotes)

Table 36: Decision-rule consistency (selected rows; Iteration 5, gen_art_evaluation_8).

Iter.	Rule	Outcome	Status
2	(a) Gate A $\rightarrow$ fallback	TRIGGERED ($0.17 < 0.50$)	MATCH
2	(b) H1 pilot-only	YES ($N_c = 38$)	MATCH
2	(c) RQ1 CONFIRMED	NOT MET	MATCH
2	(d) Held-out untouched	SEALED	MATCH
3	R1 mechanical verdict	MIXED	MATCH
3	D1 SUPPORTED on screen	NOT SUPPORTED	MATCH
4	D2 DEAD if held-out CI incl. 1	NOT DEAD (confirmed)	MATCH
4	Mechanism label	host-vocabulary	MATCH
4	G4 second-family repl.	REPLICATED	MATCH
4	R1 dead if held-out fails	R1_DEAD (Holm 0.147)	MISMATCH

Table 37: Caveats (selected rows; Iteration 5, gen_art_evaluation_8).

Test	Value	Source
CRV1 p (co-primary A_{cont})	0.0045	heldout_post.json
Wild-cluster bootstrap p	0.012	heldout_post.json
Nativeness-permutation p (500 draws)	0.026	heldout_post.json
Within-concept A-shuffle perm. p (borderline)	0.050	audit_perm.json
CRV1 rejection rate under null (anti-conservative)	17.5%	audit_perm.json
EST_bin LPM A_{cont} p (co-primary FE)	0.165	heldout_post.json
Primary FE IRR/SD ($G = 30$, inconclusive)	0.98	heldout_post.json
Screen co-primary single-paper share	87.0%	recount

Table 38: MeSH replication caveats (selected rows; Iteration 5, gen_art_evaluation_8).

Item	Value	N
Coverage rule removes share of entries	34.1%	–
Declared kw5 + coverage-0.50 design events	192	46 concepts
R1 primary FE	1.324	1,004
R2 co-primary	1.233	2,171
S3 placebo host	1.031	2,171
G1 multi-team	1.079	830

Table 39: Rooting contrasts (Iteration 5, gen_art_evaluation_8).

Type	Entries/concept	EST rate	Raw A_{cont}	CT	PPML IRR/SD	95% CI
BROAD	18.4	0.31	0.039	0.584	1.36	[1.20, 1.55]
LOCALISED	6.6	0.13	0.056	0.691	1.16	[0.98, 1.38]

Table 40: Adopter enrichment (verified values; Iteration 5, gen_art_evaluation_8).

Exposure	OR	95% CI	Note
E_any (primary, m2)	3.09	[2.32, 4.28]	prev. 0.79 vs 0.61
E_neg (frequency-matched)	0.62	[0.49, 0.77]	
E_plac (other concept's entry)	0.72	[0.57, 0.90]	
E_for (FOREIGN)	2.88	[2.28, 3.69]	
E_adj (ADJACENT)	1.91	[1.43, 2.73]	
E_nat (NATIVE)	1.47	[1.02, 2.33]	
E_any $\times$ A_{cont} interaction	0.87	[0.70, 1.14]	null

Case 1: Wireless backhaul (BROAD medoid, Computer Science, $F = 2006$). 14 host entries, only one rooted (Aerospace Engineering in 2008, $A_{\text{cont}} = 0.084$). The 13 non-rooted entries average $A_{\text{cont}} = 0.021$.

Case 2: Einstein-Podolsky-Rosen steering (BROAD, nearest non-medoid, Physics, $F = 2011$). Of its 7 co-primary entries, one is rooted: AI in 2011 ($A_{\text{cont}} = 0.142$, 37 newcomer papers). Rooted-minus-unrooted A_{cont} is +0.115.

Case 3: Locally repairable code (LOCALISED medoid, Computer Science, $F = 2013$). One co-primary entry (AI in 2013, $A_{\text{cont}} = 0.056$), rooted.

Case 4: Holographic QCD (LOCALISED, nearest non-medoid, Physics, $F = 2006$). All 5 co-primary entries are non-rooted, averaging $A_{\text{cont}} = 0.041$, with three having $\text{CT} = 1.0$: pure origin packages.

Table 41: Per-case entry statistics, co-primary sample (Iterations 3–5, `gen_art_evaluation_8`).

Concept	Entries	Rooted	Mean A_{cont} (rooted)	Mean A_{cont} (unrooted)
Wireless backhaul	14	1	0.084	0.021
EPR steering	7	1	0.142	0.027
Locally repairable code	1	1	0.056	–
Holographic QCD	5	0	–	0.041

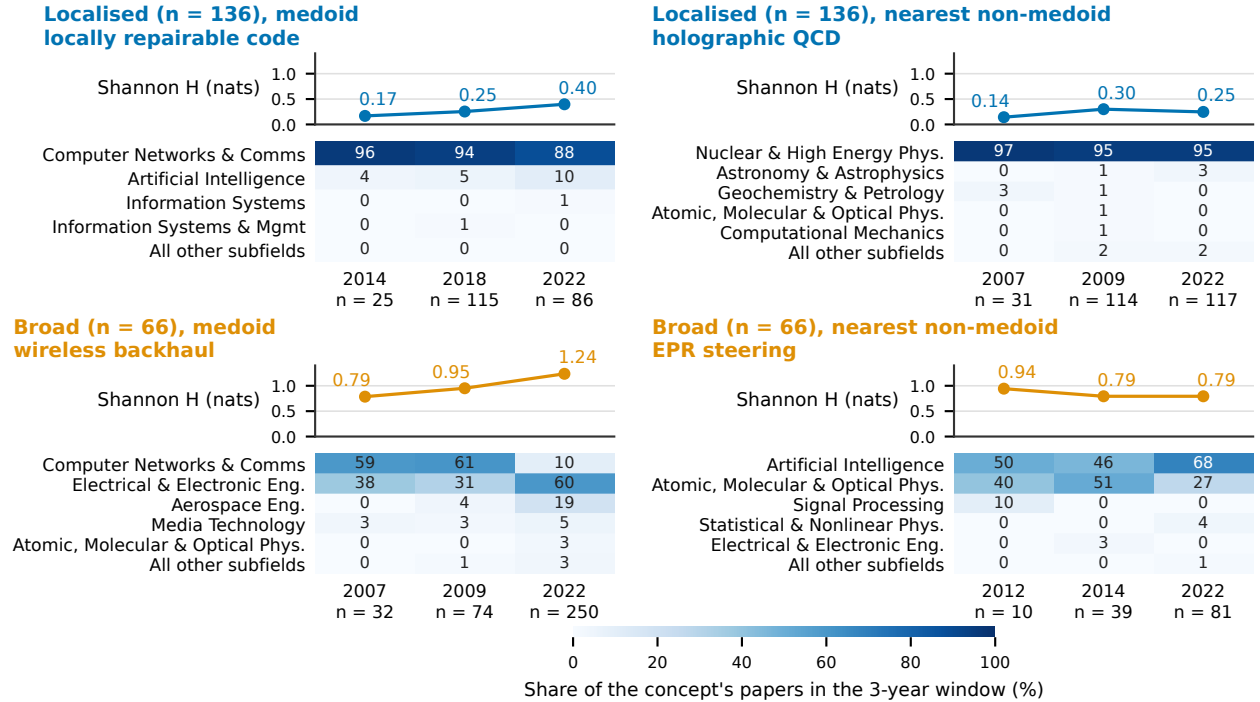


Figure 6: Representative cases of the $k = 2$ diffusion typology: localised ($n = 136$, top row) and broad-from-the-start ($n = 66$, bottom row). Each row shows the cluster’s medoid (left) and the nearest non-medoid concept (right). Localised concepts stay near-monopolised by a single subfield (entropy 0.14–0.40 nats). Broad concepts split across two or more subfields from their first window (entropy 0.79–1.24 nats).

6.5 Artifact 24: Positioning Dossier (gen_art_research_2)

Novelty verdict: PARTLY ANTICIPATED (0 ANTICIPATES, 11 PARTLY among 25 graded candidates). No located work measures how host-specific the vocabulary is that a concept is combined with when it enters a new subfield, and relates that, within concepts and across hosts, to later uptake by the host’s own newcomers.

Key correction: Cheng et al. 2023 measure global embeddedness of new ideas (outcome is yearly article COUNT, not binary “core” status). The margin held by this study is: host-specific context at entry, concept fixed effects across hosts, author-disjoint newcomer outcome, sealed-fold confirmation, and MeSH replication.

6.6 Artifact 25: Paper Figures (gen_art_evaluation_9)

Table 42: Figure inventory and production checks (Iteration 5, gen_art_evaluation_9).

Figure	Description	PDF	PNG	Values checked
F1	Method flow	Yes	Yes	39 exact
F2	D2 forest plot	Yes	Yes	123 exact, 1 derived
F3	Mechanism panel	Yes	Yes	162 exact, 1 derived
F4	RQ1 held-out	Yes	Yes	94 exact, 8 derived
F5	Occupancy-rooting	Yes	Yes	67 exact
F6	Case studies	Yes	Yes	425 exact
F6b	Case ego-networks	Yes	Yes	8 images
F7	Self-test	Yes	Yes	0 (schematic)

All 8 figures pass production checks. Total: 957 values checked, 0 failures, value fidelity 1.0.

6.7 Dead Ends and Negative Results (Iteration 5)

1. **EST_bin null on held-out.** The binary establishment threshold shows +2.54 pp per SD with $p = 0.16$.
2. **Intensive margin is conditional, not causal.**
3. **Field boundary not detected.** Wald $p = 0.638$.
4. **MeSH retention slightly lower in host-specificity test.** 12.4% of A_{cont} effect absorbed by field breadth on MeSH.
5. **Audit reveals 292 MISSING_IN_REPORT rows.**
6. **Table 18 self-test: 4 of 11 claims NOT DONE.**
7. **Seeded-perturbation recall 0.60.**

7 Run Ledger

Table 43: Run ledger: all artifacts across iterations.

Artifact	Type	Status	Iter.	OA credits
gen_art_dataset_1 (pool)	dataset	succeeded	1	2,566
gen_art_dataset_2 (background)	dataset	failed (timeout)	1	1,952
gen_art_dataset_3 (MeSH)	dataset	succeeded	1	1,784
gen_art_dataset_4 (grounding)	dataset	succeeded	1	1,847
gen_art_research_1 (dossier)	research	succeeded	1	—
gen_art_dataset_5 (hydrated)	dataset	succeeded	2	7,237
gen_art_experiment_1 (grounding)	experiment	succeeded	2	—
gen_art_experiment_2 (viability)	experiment	succeeded	2	—
gen_art_experiment_3 (emergence, main)	experiment	succeeded	2	—
gen_art_experiment_4 (emergence, MeSH)	experiment	succeeded	2	—
gen_art_evaluation_1 (audit)	evaluation	succeeded	3	—
gen_art_experiment_5 (emergence deepen)	experiment	succeeded	3	—
gen_art_experiment_6 (breadth prediction)	experiment	succeeded	3	—
gen_art_experiment_7 (host-entry grafting)	experiment	succeeded	3	—
gen_art_experiment_8 (descriptive diffusion)	experiment	succeeded	3	—
gen_art_evaluation_2 (grafting held-out)	evaluation	succeeded	4	—
gen_art_experiment_9 (MeSH replication)	experiment	succeeded	4	—
gen_art_evaluation_3 (closure held-out)	evaluation	succeeded	4	—
gen_art_evaluation_4 (descriptive held-out)	evaluation	succeeded	4	—
gen_art_evaluation_5 (adopter mechanism)	evaluation	succeeded	4	—
gen_art_evaluation_6 (margin decomposition)	evaluation	succeeded	5	—
gen_art_evaluation_7 (host-specificity)	evaluation	succeeded	5	—
gen_art_evaluation_8 (full audit)	evaluation	succeeded	5	—
gen_art_research_2 (positioning dossier)	research	succeeded	5	—
gen_art_evaluation_9 (paper figures)	evaluation	succeeded	5	—

8 Coverage Against the Original Request

Table 44: Final coverage against the original request.

#	Activity	Status and caveats
1	Prepare and semantically ground dataset	Done. Grounding covers focal concepts only; co-word network uses $\sim 27k$ legacy OpenAlex nodes.
2	Construct evolving knowledge network	Done. Legacy-concept nodes; variant normalisation minimal.
3	RQ1: temporal network analysis	Done. R1_DEAD on primary; persistent-neighbour confirmed.
4	RQ2: cross-disciplinary diffusion	Done. Grafting confirmed + MeSH replicated + K1/K2/K3 decomposition. Primary spec underpowered; Physics null within sampling variation; EST_bin null on held-out; intensive margin conditional.
5	Identify recurring trajectories	Done. $k = 2$ replicates on held-out and MeSH. Finer typologies unstable; 3-channel adds nothing beyond entropy.
6	Validate with representative cases	Done. Four medoid cases with ego-network visualisations and alluvial paths.

9 What the Run Learned Overall and What Is Still Open

9.1 Confirmed Findings

Host-entry grafting predicts durable integration. When a concept enters a new host subfield, the share of its initial co-occurrence partners that are native to that host predicts subsequent uptake by newcomer authors. This holds on the held-out fold (co-primary IRR/SD 1.19 [1.06, 1.33], $p = 0.004$) and replicates on the independent MeSH population (IRR/SD 1.23 [1.12, 1.36], Holm $p < 0.001$). The IVW pooled estimate is 1.26 [1.17, 1.36] with $I^2 = 0$. Co-transfer (arriving with origin companions) is null across all folds and populations.

The effect operates through both the extensive margin (IVW +6.43 pp/SD [4.76, 8.11]) and the intensive margin (IVW IRR/SD 1.183 [1.104, 1.267]), is host-specific rather than reflecting general accessibility (IVW retention 0.969 after generality controls; Balassa-index A_{lift} IVW IRR/SD 1.34 [1.23, 1.46]), and does not vary by origin field (Wald $p = 0.638$).

Persistent-neighbour closure is confirmed. Among stable top-20 associates, emerging concepts show lower triadic closure before onset (held-out $S = -1.073$, Holm $p = 0.015$). Burt constraint is positive on both folds, ruling out a simple brokerage interpretation.

Expansion precedes diffusion. In 25 of 26 pooled main-arm dual-onset concepts, network expansion precedes disciplinary diffusion.

9.2 Dead Findings

- **R1_DEAD:** The general closure deficit did not survive held-out confirmation on the primary operationalisation (Holm $p = 0.147$).
- **Source-sink viability:** Blocked by Gate A failure (within-host traced share 0.17).
- **Breadth prediction:** Openness does not predict where concepts travel beyond growth base-lines.
- **Closure-anchoring link:** Null. The two findings (closure and grafting) are independent.
- **Co-transfer:** Null across all specifications and populations.

9.3 Adopter-Level Mechanism

Adopters are 3.1 times more likely than matched non-adopters to have prior exposure to the concept’s entry partners (OR 3.09 [2.32, 4.28]), with enrichment strongest for origin-vocabulary (FOREIGN) partners (OR 2.88). This is consistent with absorptive capacity or topical proximity. The vocabulary-class gradient (FOREIGN > NATIVE) runs opposite to the entry-level grafting finding, indicating that the two levels operate through different channels: at the entry level, host-native partners lower the barrier for newcomers who lack origin expertise; at the adopter level, those who adopt tend to have origin expertise already.

9.4 What Is Still Open

1. **Generalisation beyond physics/CS/biomedicine.** The population is dominated by arXiv-sourced concepts. Social sciences and humanities are untested.
2. **Binary establishment threshold.** EST_bin is null on the held-out fold ($p = 0.165$). The effect predicts uptake initiation, not strict establishment.
3. **Primary specification.** The fully saturated primary specification remains inconclusive on the held-out fold (IRR/SD 0.98, $G = 30$).
4. **Single-paper caveat.** 87% of co-primary events are single-paper entries. On the held-out fold, the single-paper interaction ratio is 0.80 ($p = 0.004$).
5. **Causal identification.** The study design is observational. Host-nativeness of entry partners is not randomly assigned.
6. **Finer diffusion typologies.** Only $k = 2$ is stable; finer typologies are unstable.
7. **The absorptive-capacity vs topical-proximity distinction** at the adopter level cannot be resolved with this design.

9.5 Pre-Registration Integrity

All pre-registered specifications were frozen and hashed before outcome data were inspected. Fourteen specifications were hashed in total, covering the concept frame, viability rules, test population, turnover pre-check, emergence deepen, host-entry grafting, descriptive diffusion held-out, held-out confirmation, grafting held-out, discriminating tests, MeSH replication, closure-anchoring link, and the K1/K2/K3 decomposition. Hash digests are recorded in the run’s artifact metadata and can be verified against the original specification files.